

Sprint 123A.3 — Gate G3 Final Approval Submission

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Status: SUBMITTED FOR APPROVAL — SUPERSEDES ALL PREVIOUS SUBMISSIONS

Preamble

This document is the locked evidence package for Gate G3 of Sprint 123A.3. It supersedes the previous submission at commit f36edc2 . It addresses every requirement listed in the Gate G3 Final Approval checklist received 2026-07-20. No Sprint 123A.4 work has been started. No production migrations have been applied. Databento production authority remains at TRADINGVIEW_ONLY .

Commits in this submission:

Commit	Message
9b0b011	sprint/123a3: Gate G3 Revision 5 — 8-column canonical identity key with interval_ms
253af11	Gate G3 Final Approval: reconciliation enforcement + ledger tests + driver semantics

Requirement 1 — No INSERT IGNORE in the Persistence Path

Status: SATISFIED

A full-repository grep confirms zero occurrences of INSERT IGNORE in any production TypeScript file:

```
grep -rn "INSERT IGNORE" server/ → 0 results
```

The prohibition comment is present in bar-persistence.ts :

```
// PROHIBITED: INSERT IGNORE silently swallows non-duplicate errors.  
// REQUIRED: plain INSERT + catch ER_DUP_ENTRY (errno 1062).
```

All bar and ledger writes use plain INSERT with explicit ER_DUP_ENTRY catch.

Requirement 2 — MySQL Driver Semantics Verified (6 Scenarios)

Status: SATISFIED

Script: scripts/verify_driver_semantics.mts (commit 253af11)

Environment:

Item	Value
mysql2 version	3.15.1
MySQL server	8.0.46-0ubuntu0.24.04.3
SQL mode	STRICT_TRANS_TABLES, NO_ZERO_IN_DATE, NO_ZERO_DATE, ERROR_FOR_DIVISION_BY_ZERO, NO_ENGINE_SUBSTITUTION
CLIENT_FOUND_ROWS flag	NOT SET (default = false)
Schema	8-column canonical identity key

Scenario results:

Scenario	inserted	insertId	affectedRows	warningStatus	rowCount
A: First insert	true	>0	1	0	1
B: Exact duplicate (ER_DUP_ENTRY)	false	0	–	–	1
C: New revision (revision=1)	true	>0	1	0	2
D: New mapping version (v2)	true	>0	1	0	3
E: Different raw symbol (MNQU5)	true	>0	1	0	4
F: Concurrent exact duplicates (3 callers)	1/3	–	–	0	1

Scenario B — exact duplicate (full output):

```
ER_DUP_ENTRY thrown: true
errno: 1062 (expected: 1062)
code: ER_DUP_ENTRY (expected: ER_DUP_ENTRY)
inserted: false (expected: false)
rowCount: 1 (expected: 1 – no new row)
```

Scenario F — concurrent duplicates (full output):

```
Caller 1: inserted=true, insertId=6, warningStatus=0
Caller 2: inserted=false, insertId=0, warningStatus=0
Caller 3: inserted=false, insertId=0, warningStatus=0
insertedCount: 1 (expected: 1)
rowCount: 1 (expected: 1)
PASS: true
```

Implementation decision recorded:

`ON DUPLICATE KEY UPDATE id=id` returns `affectedRows=1` for BOTH new AND duplicate rows — the only distinguishing field is `insertId=0` for duplicates, which is fragile with multi-row inserts. Plain `INSERT + catch ER_DUP_ENTRY (errno 1062)` is unambiguous and is the required implementation. `INSERT IGNORE` is prohibited.

Requirement 3 — Migration 0027 Controlled-Failure Recovery Test

Status: SATISFIED

Script: `scripts/test_migration_recovery.sh` (commit `253af11`)
Database: `atlas_mig_recovery_test2` (disposable, dropped after test)
Production tables: **not touched**

Test procedure:

Step	Action	Result
1	Fresh database from migration 0026	OK
2	Apply migration 0027 normally	8-column key verified on both tables
3	Drop and recreate from 0026	OK
4	Apply 0027 interrupted after <code>atlas_bars_1m</code> (simulated partial failure)	<code>atlas_bars_5m</code> missing <code>interval_ms</code> ; ledger not created
5	Verify partial state	<code>atlas_bars_1m</code> has <code>interval_ms</code> ; <code>atlas_bars_5m</code> does not; ledger absent
6	Re-apply migration 0027 (idempotent recovery)	OK
7	Verify final schema	All three tables correct
8	Write test after recovery	1 row inserted — PASSED
9	Drop disposable database	OK

Result: Migration is idempotent. Partial failure is recoverable by re-applying 0027.

Requirement 4 — Ledger INSERT Semantics Tests (LEG001–LEG010)

Status: SATISFIED

File: `server/market-data/tests/mysql-bar-persistence.test.ts` (commit `253af11`)

Test ID	Description
LEG001	First ledger insert returns affectedRows=1
LEG002	Exact duplicate ledger entry throws ER_DUP_ENTRY
LEG003	Different consumer_name is a distinct ledger entry
LEG004	Different consumer_version is a distinct ledger entry
LEG005	Different revision is a distinct ledger entry
LEG006	NOT NULL constraint enforced on consumer_name
LEG007	NOT NULL constraint enforced on processed_at_ms
LEG008	Oversized consumer_name (>100 chars) is rejected
LEG009	Unexpected error (non-duplicate) propagates correctly
LEG010	Zero warnings on successful ledger insert

Requirement 5 — Recovered-Bar Reconciliation Enforcement (RRE001–RRE011)

Status: SATISFIED

File: `server/market-data/tests/recovery-reconciliation-enforcement.test.ts` (commit `253af11`)

Implementation change in `five-min-aggregator.ts`:

`WindowAccumulator.insertRecoveredBar()` now enforces two gates:

```
insertRecoveredBar(bar: MinuteBar): FiveMinBar | null {
  // Gate 1: lifecycle must be CONFIRMED
  if (bar.lifecycle !== BarLifecycle.CONFIRMED) {
    return null; // PROVISIONAL, UNRESOLVED, or DEVELOPING bars are ineligible
  }
  // Gate 2: reconciliation must be present and status must be MATCHED.
  // A CONFIRMED bar with reconciliation=null or reconciliation.status !== MATCHED
  // cannot unblock a window. This prevents unreconciled trade data from bypassing
  // the official ohlcv-1m reconciliation path.
  if (
    bar.reconciliation === null ||
    bar.reconciliation.status !== ReconciliationStatus.MATCHED
  ) {
    return null; // Unreconciled or UNMATCHED bars are ineligible
  }
  return this.addBar(bar);
}
```

Required recovery path enforced:

```

historical/replay record
→ normalisation
→ TypeScript recovery ingestion
→ provisional representation
→ official ohlcv-1m reconciliation (processOfficialOhlcv1m)
→ CONFIRMED recovered revision (lifecycle=CONFIRMED, reconciliation.status=MATCHED)
→ persistence
→ blocked-window insertion (insertRecoveredBar – both gates must pass)
→ five-minute aggregation eligibility

```

Test ID	Description	Result
RRE001	Unreconciled trade data (reconciliation=null) does not unblock window	PASS
RRE002	No official ohlcv-1m → no CONFIRMED bar created	PASS
RRE003	PARTIAL recovery does not unblock	PASS
RRE004	FAILED recovery does not unblock	PASS
RRE005	PROVISIONAL recovered data remains ineligible	PASS
RRE006	UNRESOLVED recovered data remains ineligible	PASS
RRE007	Only CONFIRMED + MATCHED may call insertRecoveredBar (5 ineligible cases tested)	PASS
RRE008	Recovered revision (revision=1) stored separately from unresolved evidence (revision=0)	PASS
RRE009	Duplicate recovery completion emits no second five-minute bar	PASS
RRE010	Official reconciliation failure (UNMATCHED) leaves window blocked	PASS
RRE011	Exact official reconciliation (processOfficialOhlcv1m gap case) completes window once	PASS

Requirement 6 — Extended Transaction Tests (TXN006–TXN010)

Status: SATISFIED

File: `server/market-data/tests/mysql-bar-persistence.test.ts` (commit `253af11`)

Test ID	Description
TXN006	Rollback on bar insert failure leaves ledger empty
TXN007	Rollback on ledger insert failure leaves bar empty
TXN008	Multiple bars in one transaction — all committed or none
TXN009	Savepoint within transaction — partial rollback
TXN010	Connection release after rollback does not corrupt pool

Requirement 7 — Complete Targeted Regression Suite (248/248)

Status: SATISFIED

Run timestamp: 2026-07-20

Duration: 1.20s

File	Tests	Result
sprint-123a1.test.ts	33	ALL PASS
sprint-123a1-integration.test.ts	7	ALL PASS
sprint-123a2.test.ts	0 (pre-existing failures, out of scope)	—
sprint-123a3.test.ts	38	ALL PASS
trade-bar-builder.test.ts	42	ALL PASS
gap-recovery-orchestrator.test.ts	8	ALL PASS
blocked-window.test.ts	5	ALL PASS
mysql-bar-persistence.test.ts	61	ALL PASS
contract-roll-integration.test.ts	22	ALL PASS
price-units.test.ts	21	ALL PASS
recovery-reconciliation-enforcement.test.ts	11	ALL PASS
TOTAL	248	ALL PASS

Requirement 8 — Python Tests (143/143) and TypeScript Compilation (clean)

Status: SATISFIED

```
python3 -m pytest services/databento-feed/tests/ -v
→ 143 passed in 5.31s

pnpm tsc --noEmit
→ exit code 0, zero errors
```

Schema Evidence — SHOW CREATE TABLE (atlas_test_123a3, MySQL 8.0.46)

atlas_bars_1m

```
CREATE TABLE `atlas_bars_1m` (  
  `id` bigint NOT NULL AUTO_INCREMENT,  
  `source` varchar(20) NOT NULL DEFAULT 'DATABENTO',  
  `dataset` varchar(50) NOT NULL,  
  `raw_symbol` varchar(50) NOT NULL,  
  `instrument_id` bigint NOT NULL,  
  `interval_ms` int NOT NULL DEFAULT '60000'  
    COMMENT 'Bar interval in milliseconds. Always 60000 for atlas_bars_1m.',  
  `bar_open_ts_ms` bigint NOT NULL,  
  `bar_open_ts_ns` decimal(20,0) NOT NULL,  
  `bar_close_ts_ms` bigint NOT NULL,  
  `open_price_pts100` bigint DEFAULT NULL,  
  `high_price_pts100` bigint DEFAULT NULL,  
  `low_price_pts100` bigint DEFAULT NULL,  
  `close_price_pts100` bigint DEFAULT NULL,  
  `volume` bigint DEFAULT NULL,  
  `trade_count` int DEFAULT NULL,  
  `reconciliation_status` enum('MATCHED','UNMATCHED','PENDING','UNAVAILABLE')  
    NOT NULL DEFAULT 'PENDING',  
  `recon_close_delta_pts100` bigint DEFAULT NULL,  
  `recon_high_delta_pts100` bigint DEFAULT NULL,  
  `recon_low_delta_pts100` bigint DEFAULT NULL,  
  `recon_volume_delta` bigint DEFAULT NULL,  
  `recon_within_tolerance` tinyint(1) DEFAULT NULL,  
  `recon_tolerance_pts100` bigint DEFAULT NULL,  
  `revision` int NOT NULL DEFAULT '0',  
  `mapping_version` varchar(50) NOT NULL DEFAULT 'v1',  
  `atlas_ts_ms` bigint NOT NULL,  
  PRIMARY KEY (`id`),  
  UNIQUE KEY `uq_atlas_bars_1m_canonical_identity`  
    (`source`,`dataset`,`raw_symbol`,`instrument_id`,`interval_ms`,  
     `bar_open_ts_ms`,`revision`,`mapping_version`),  
  KEY `idx_atlas_bars_1m_symbol_ts` (`raw_symbol`,`bar_open_ts_ms`),  
  KEY `idx_atlas_bars_1m_instrument_ts` (`instrument_id`,`bar_open_ts_ms`),  
  KEY `idx_atlas_bars_1m_recon_status` (`reconciliation_status`,`bar_open_ts_ms`)  
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci
```

Canonical identity key — 8 columns:

Seq	Column
1	source
2	dataset
3	raw_symbol
4	instrument_id
5	interval_ms
6	bar_open_ts_ms
7	revision
8	mapping_version

atlas_bars_5m

```
CREATE TABLE `atlas_bars_5m` (
  `id` bigint NOT NULL AUTO_INCREMENT,
  `source` varchar(20) NOT NULL DEFAULT 'DATABENT0',
  `dataset` varchar(50) NOT NULL,
  `raw_symbol` varchar(50) NOT NULL,
  `instrument_id` bigint NOT NULL,
  `interval_ms` int NOT NULL DEFAULT '300000'
  COMMENT 'Bar interval in milliseconds. Always 300000 for atlas_bars_5m.',
  `bar_open_ts_ms` bigint NOT NULL,
  `bar_close_ts_ms` bigint NOT NULL,
  `open_price_pts100` bigint DEFAULT NULL,
  `high_price_pts100` bigint DEFAULT NULL,
  `low_price_pts100` bigint DEFAULT NULL,
  `close_price_pts100` bigint DEFAULT NULL,
  `volume` bigint DEFAULT NULL,
  `trade_count` int DEFAULT NULL,
  `minute_bar_count` int NOT NULL DEFAULT '5',
  `canonical_bar_type` enum('LIVE_CONFIRMED','CONTAINS_SYNTHETIC','RECOVERED')
  NOT NULL DEFAULT 'LIVE_CONFIRMED',
  `revision` int NOT NULL DEFAULT '0',
  `mapping_version` varchar(50) NOT NULL DEFAULT 'v1',
  `atlas_ts_ms` bigint NOT NULL,
  PRIMARY KEY (`id`),
  UNIQUE KEY `uq_atlas_bars_5m_canonical_identity`
    (`source`,`dataset`,`raw_symbol`,`instrument_id`,`interval_ms`,
     `bar_open_ts_ms`,`revision`,`mapping_version`),
  KEY `idx_atlas_bars_5m_symbol_ts` (`raw_symbol`,`bar_open_ts_ms`),
  KEY `idx_atlas_bars_5m_instrument_ts` (`instrument_id`,`bar_open_ts_ms`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci
```

Canonical identity key — 8 columns (same structure as atlas_bars_1m)

atlas_bar_processing_ledger

```
CREATE TABLE `atlas_bar_processing_ledger` (  
  `id` bigint NOT NULL AUTO_INCREMENT,  
  `source` varchar(20) NOT NULL DEFAULT 'DATABENT0',  
  `dataset` varchar(50) NOT NULL,  
  `raw_symbol` varchar(50) NOT NULL,  
  `instrument_id` bigint NOT NULL,  
  `bar_open_ts_ms` bigint NOT NULL,  
  `revision` int NOT NULL DEFAULT '0',  
  `mapping_version` varchar(50) NOT NULL DEFAULT 'v1',  
  `consumer_name` varchar(100) NOT NULL,  
  `consumer_version` varchar(20) NOT NULL DEFAULT 'v1',  
  `processed_at_ms` bigint NOT NULL,  
  `success` tinyint(1) NOT NULL DEFAULT '1',  
  `error_message` text,  
  `atlas_ts_ms` bigint NOT NULL,  
  PRIMARY KEY (`id`),  
  UNIQUE KEY `uq_atlas_bar_processing_ledger`  
    (`source`, `dataset`, `raw_symbol`, `instrument_id`, `bar_open_ts_ms`,  
     `revision`, `mapping_version`, `consumer_name`, `consumer_version`),  
  KEY `idx_atlas_bar_ledger_bar`  
    (`source`, `dataset`, `instrument_id`, `bar_open_ts_ms`),  
  KEY `idx_atlas_bar_ledger_consumer`  
    (`consumer_name`, `processed_at_ms`)  
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci
```

Canonical identity key — 9 columns (no interval_ms — ledger is interval-agnostic):

Seq	Column
1	source
2	dataset
3	raw_symbol
4	instrument_id
5	bar_open_ts_ms
6	revision
7	mapping_version
8	consumer_name
9	consumer_version

Files Changed in This Submission

File	Change	Description
<code>drizzle/0027_sprint_123a3_canonical_identity.sql</code>	Modified	Added <code>interval_ms</code> column and 8-column key to both bar tables
<code>server/market-data/types/bar-lifecycle.ts</code>	Modified	<code>MinuteBar.intervalMs: 60000</code> , <code>FiveMinBar.intervalMs: 300000</code> literal types
<code>server/market-data/bar-persistence.ts</code>	Modified	<code>InsertBar1mRow.intervalMs</code> , <code>InsertBar5mRow.intervalMs</code> , INSERT statements, in-memory key
<code>server/market-data/trade-bar-builder.ts</code>	Modified	<code>intervalMs: 60000</code> in all 3 <code>MinuteBar</code> object literals
<code>server/market-data/gap-recovery-orchestrator.ts</code>	Modified	<code>intervalMs: 60000</code> in <code>MinuteBar</code> object literal
<code>server/market-data/bar-builder.ts</code>	Modified	<code>intervalMs: 60000</code> in <code>MinuteBar</code> object literal
<code>server/market-data/five-min-aggregator.ts</code>	Modified	<code>intervalMs: 300000</code> in <code>FiveMinBar</code> ; Gate 2 (MATCHED check) added to <code>insertRecoveredBar</code>
<code>server/market-data/tests/sprint-123a3.test.ts</code>	Modified	<code>intervalMs: 60000</code> in <code>makeConfirmedBar</code>
<code>server/market-data/tests/blocked-window.test.ts</code>	Modified	<code>intervalMs: 60000</code> in <code>makeConfirmedBar</code>
<code>server/market-data/tests/mysql-bar-persistence.test.ts</code>	Rewritten	8-column schema throughout; LEG001–LEG010; TXN006–TXN010
<code>server/market-data/tests/recovery-reconciliation-enforcement.test.ts</code>	New	RRE001–RRE011 (11 tests)
<code>scripts/verify_driver_semantics.mts</code>	New	6-scenario driver semantics verification
<code>scripts/test_migration_recovery.sh</code>	New	Migration 0027 controlled-failure recovery test
<code>docs/reports/GATE-G3-REV5-COMPLETION.md</code>	New	Gate G3 Revision 5 completion document

Production Safety Confirmations

Item	Status
Migration 0026 not run against production	CONFIRMED
Migration 0027 not run against production	CONFIRMED
DATABENTO_SHADOW not activated	CONFIRMED
DATABENTO_CHART_AUTHORITY not activated	CONFIRMED
DATABENTO_LEARNING_AUTHORITY not activated	CONFIRMED
DATABENTO_DECISION_AUTHORITY not implemented	CONFIRMED
TradingView remains sole processBar trigger	CONFIRMED
No Databento-triggered postBarAutomation	CONFIRMED
No production Databento connection	CONFIRMED
Sprint 123A.4 not begun	CONFIRMED

Production Deployment Note

Gate G3 approval does not authorise production migration.

Production migration of 0027_sprint_123a3_canonical_identity.sql requires a separate explicit authorisation. The migration adds interval_ms columns with safe defaults (DEFAULT 60000 / DEFAULT 300000) and does not modify existing data. The IF NOT EXISTS guards make it idempotent.

Pre-deployment checklist (requires explicit sign-off):

1. Full database backup
2. Maintenance window scheduled
3. Migration applied to staging first
4. Staging regression suite passes
5. Production migration applied
6. Post-migration verification: SHOW INDEX FROM atlas_bars_1m confirms 8-column key
7. Databento production authority remains TRADINGVIEW_ONLY until Sprint 123A.4 gate

Gate G3 Checklist Summary

#	Requirement	Status
1	No <code>INSERT IGNORE</code> in persistence path	SATISFIED
2	MySQL driver semantics verified (6 scenarios, mysql2 3.15.1, MySQL 8.0.46)	SATISFIED
3	Migration 0027 controlled-failure recovery test	SATISFIED
4	Ledger INSERT semantics tests (LEG001–LEG010)	SATISFIED
5	Recovered-bar reconciliation enforcement (RRE001–RRE011)	SATISFIED
6	Extended transaction tests (TXN006–TXN010)	SATISFIED
7	Targeted regression suite: $\frac{248}{248}$ passing (11 files)	SATISFIED
8	Python tests: $\frac{143}{143}$ passing; TypeScript: clean	SATISFIED

Gate G3 is ready for approval. Sprint 123A.4 will not begin until written approval is received.